

plt_sim version 3.2

JLT, SAA

2025-06-27

History

- From plt_sim version 3.1 2025-06-23. Uses spawners_m8_beta.stan
- From plt_sim_v3.qmd 2025-06-23. This and prev use spawners_m8.stan
- From plt_sim_v2.qmd 2025-06-14
- From plt_sim.R 2025-05-29

Simulation

Goal: Automate multiple simulations to demonstrate annual values of *run size* and spawning parameters (*arrival_timing*, *arrival_spread*, *exit_lag*, *exit_spread*) are recovered by fitting model spawner_m8.stan. All four spawning params as groups (multi-level regression) of annual values.

Simulation parameters

Assign distributions to parameters for runs and spawner behaviour. Set years and days for surveys. Params logged or not as per JLT code: spawners_m8.stan

```
log_run_mu           <- 10    # log value for lognormal PDD,
log_run_sigma        <- 1     # log value for lognormal PDD,
arrival_timing_mu     <- 35    # normal
arrival_timing_sigma  <- 3     # 95% days 29 to 41
arrival_spread_mu     <- 4     # normal
arrival_spread_sigma  <- 1     # 95% 2 to 6
exit_lag_mu          <- 10    # normal
exit_lag_sigma       <- 2     # 95% 6 to 14
exit_spread_mu       <- 3     # normal. should be < arrival_spread_mu
```

```

exit_spread_sigma      <- 0.66 # 95% 1.68 to 4.32

years_n                <- 12   # number of years, 4x3 plot
days_n                <- 100  # number of days, checked below
pct_surveyed           <- 0.8  # fraction of days sampled
samp_accuracy          <- 0.1  # rnorm(n,1,0.1) 10% sampling error
live_phi               <- 10   # sampling error (version 3.1)

```

Range of days with significant spawners conceivably present: 20 to 75

Draws for annual values

Draw random samples for annual runs and spawning params. Spawners present each day is cumulative arrived minus cumulative exited based on those draws.

```

run 99%CL: 2151 225561
     draws: 13341 35802 63289 34544 28987 6608 134578 18010 51598 11127 432096 51056

arrival_timing 99%CL: 28.02 41.98
               draws: 31.61 33.17 37.52 39.4 37.97 36.98 37.26 32.55 41.11 34.74 32.01 36.16

arrival_spread 99%CL: 1.67 6.33
                 draws: 4.22 4.92 2.79 3.93 2.93 4.2 2.65 3.1 4.63 3.67 5.23 3.47

exit_lag 99%CL: 5.35 14.65
           draws: 8.67 10.89 7.07 9.42 11.03 7.1 9.32 8.75 10.7 12.7 7.53 10.31

exit_spread 99%CL: 1.46 4.54
              draws: 3.37 3.19 3.61 3.82 2.62 4.17 2.13 2.47 1.69 4.09 3.03 3.73

```

Table 1: Typically, delta_spread, as arrival_spread - exit_spread, is positive. If negative, and with small exit_lag, this can lead to salmon exiting before arriving. Checked as lag between day 95% arrived to day 95% exited.

Arrival_Spread	Exit_Spread	Difference	Exit_lag	Days_95_In_Before_95_Out
4.22	3.37	0.85	8.67	7.3
4.92	3.19	1.73	10.89	8.0

Arrival_Spread	Exit_Spread	Difference	Exit_lag	Days_95_In_Before_95_Out
2.79	3.61	-0.82	7.07	8.4
3.93	3.82	0.11	9.42	9.2
2.93	2.62	0.31	11.03	10.5
4.20	4.17	0.03	7.10	7.1
2.65	2.13	0.52	9.32	8.5
3.10	2.47	0.63	8.75	7.7
4.63	1.69	2.94	10.70	5.9
3.67	4.09	-0.42	12.70	13.4
5.23	3.03	2.20	7.53	3.9
3.47	3.73	-0.26	10.31	10.7

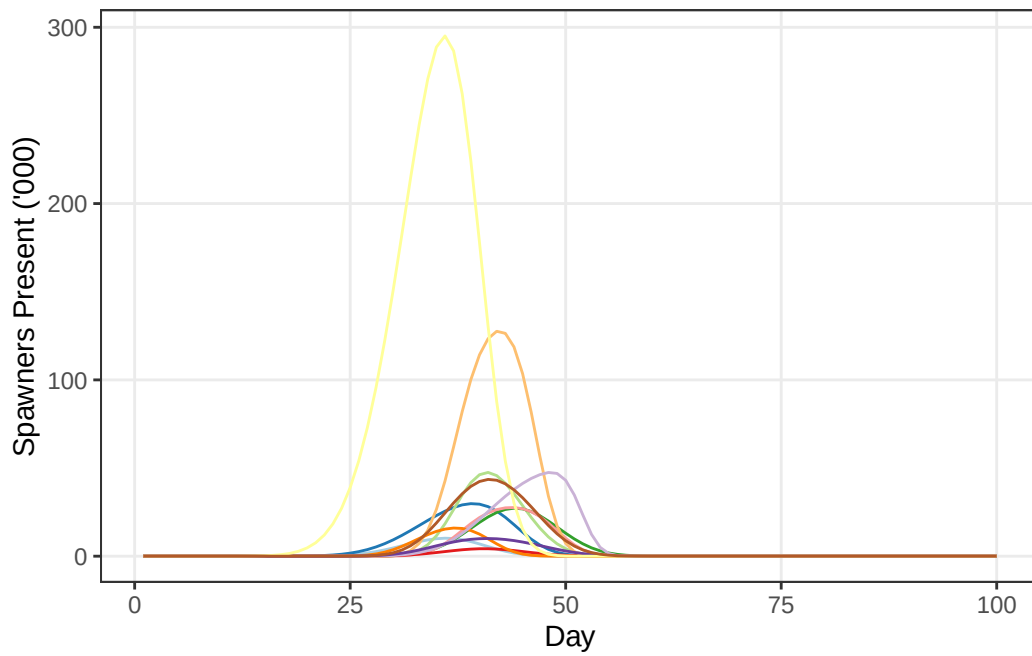


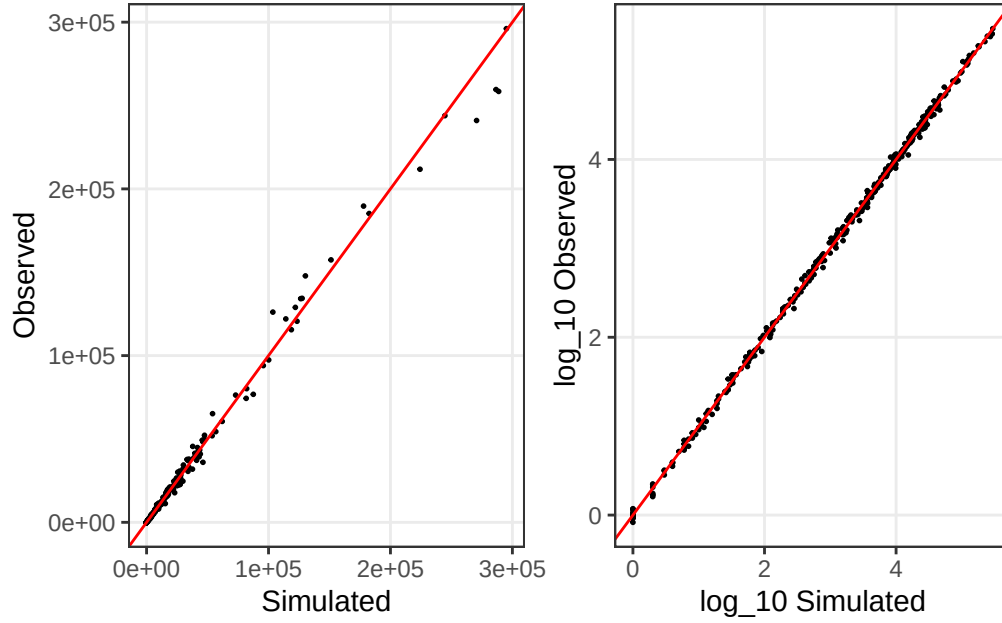
Figure 1: Time series of spawners present in each year.

Surveys

Define surveys as a random fraction of all days. Add negative binomial sampling error. Remove simulated values of zero and sampled observations of zero.

max gap between surveys each year:

[1] 3 2 3 2 1 4 3 1 2 2 1 1



(a) Sampling error is proportional, % accuracy. Note $\log(0) = 1$.

Figure 2: Comparison of spawner observed to spawners present. Note $\log(0) = 1$.

Observed time series

Plotted as observed to show scatter near peak (mode) day and as logs to better show negative skew when *arrival_spread* is greater than *exit_spread* (typical), and positive skew when smaller.

Fit

Priors for Stan

Bayesian priors affect Markov Chain Monte Carlo (or related) sampling of the parameters hyperspace. The spawning parameters: *arrival_timing*, *arrival_spread*, *exit_lag*, *exit_spread* are treated as multilevel, groups of annual values. Priors for a group define an initial distribution for the annual values: $exit_lag = \Phi(\mu, \sigma)$ has group parameters *exit_lag_mu* and *exit_lag_sigma*. Annual values are sampled (fitted) from the group distribution, and group parameters are simultaneously fitted, thereby “sharing information” to improve the fit for each annual value. Note *log_runs* is not a group, the same prior applies to each year.

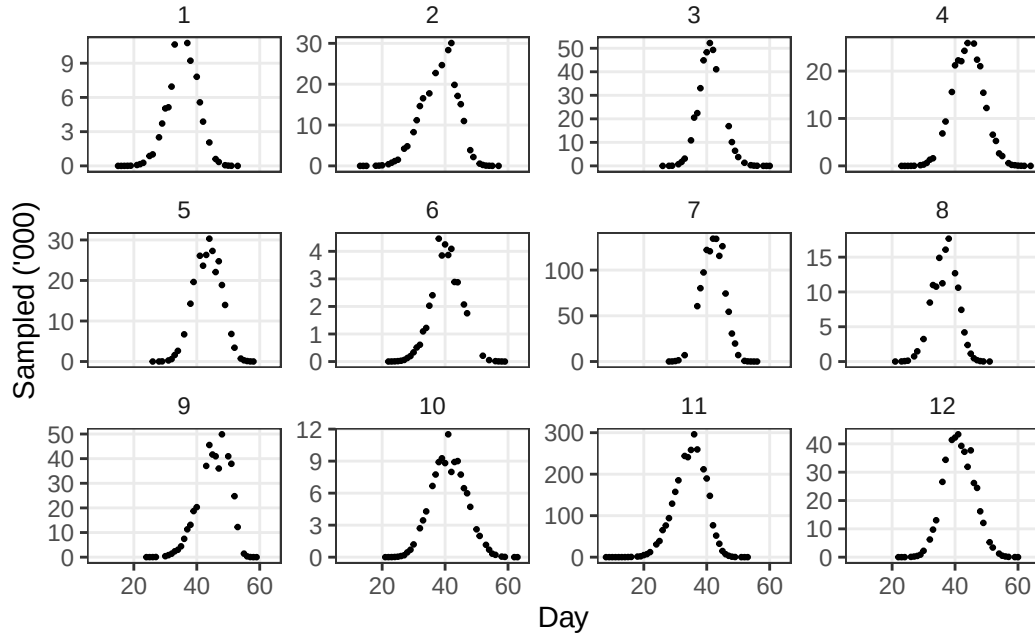
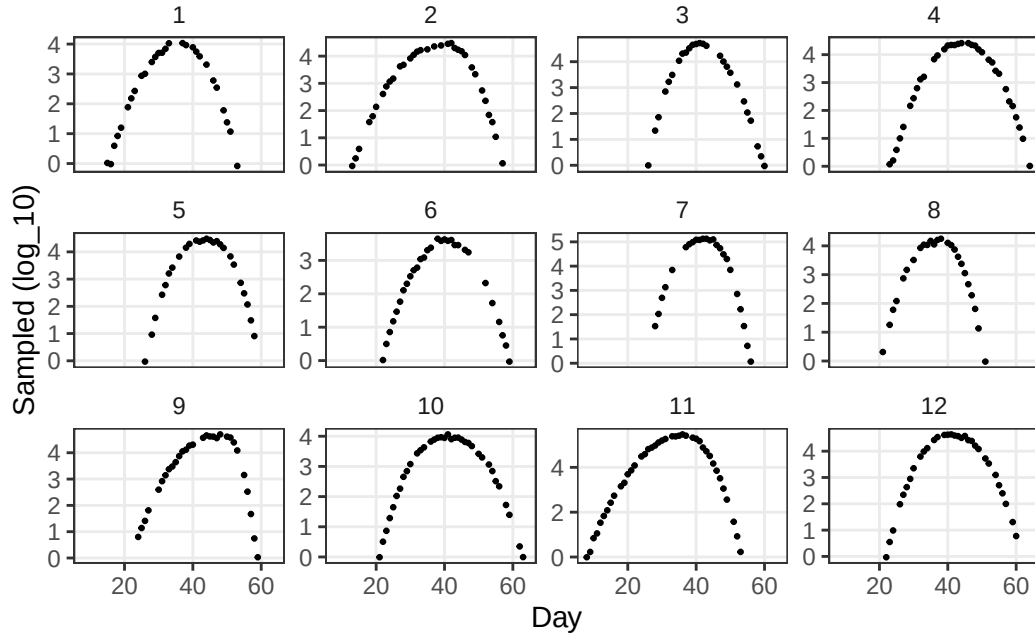


Figure 3: Annual time series of spawners observed.



(a) Reveals skewed distributions.

Figure 4: Annual time series, logged, of spawners observed.

In what follows, the prior *exit_lag_sigma* uses the value for *spread_sigma*. The Stan code renames *count_dispersion* as *log(live_phi)* where *live_phi* is the size parameter for the negative binomial distribution.

Table 2: Priors for Stan sampling of parameters space.

prior	pdd	v1	v2
log_runs_mu	lnorm	10.00	1.0
arrival_mu	norm	35.00	5.0
arrival_sigma	gamma	6.00	2.0
spread_mu	lnorm	1.95	0.3
spread_sigma	gamma	2.00	1.0
exit_lag_mu	lnorm	2.30	0.3
count_dispersion	lnorm	2.00	0.5

Data list for Stan

The priors are part of the Stan data list. Convergence of the HMC sampling chains is improved by providing initial values for *arrival_mu*. .

```
sp_dat = list(
  n_priors      = nrow(priors),
  priors       = data.matrix(priors[,3:4]),
  n_years      = years_n,
  n_obs        = nrow(sampled.df2),
  year         = sampled.df2$year,
  day          = sampled.df2$day,
  live_counts  = as.integer(sampled.df2$sampled),
  live_phi     = 10,          # rnbinom_size
  arrival_mu   = arrival_timing_mu ) # 35
```

Stan sampling

Warning: There were 275 transitions after warmup that exceeded the maximum treedepth. Increase the maximum treedepth. See <https://mc-stan.org/misc/warnings.html#maximum-treedepth-exceeded>

Warning: Examine the pairs() plot to diagnose sampling problems

Warning: Bulk Effective Samples Size (ESS) is too low, indicating posterior means and medians may be unreliable. Running the chains for more iterations may help. See <https://mc-stan.org/misc/warnings.html#bulk-ess>

Warning: Tail Effective Samples Size (ESS) is too low, indicating posterior variances and tails are poorly estimated.
Running the chains for more iterations may help. See
<https://mc-stan.org/misc/warnings.html#tail-ess>

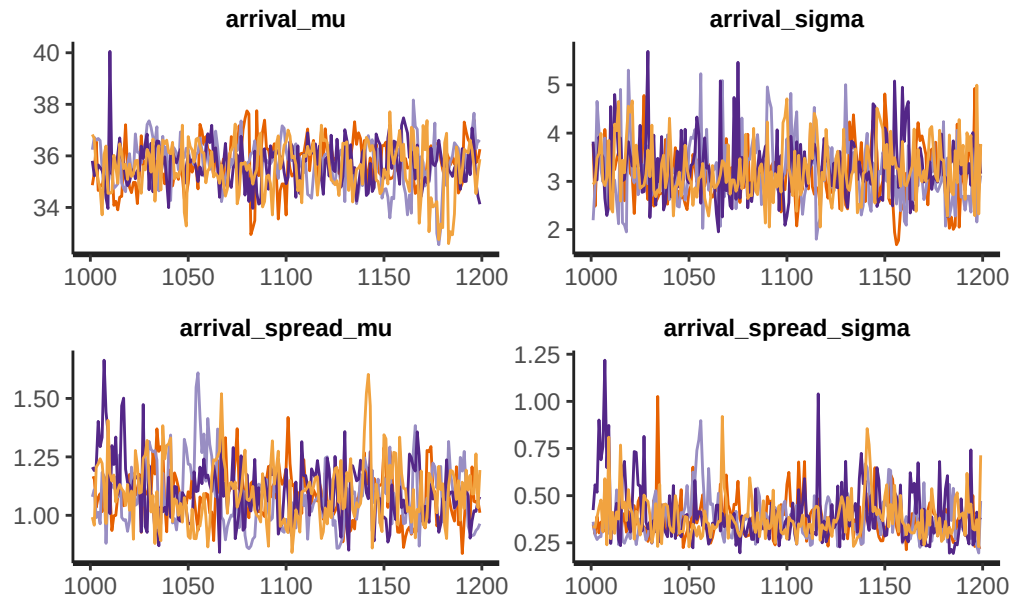


Figure 5

Fitted runs

Spawner abundance estimates presented logged and arithmetic plots. $\log_{10}/\log_e = 0.4343$

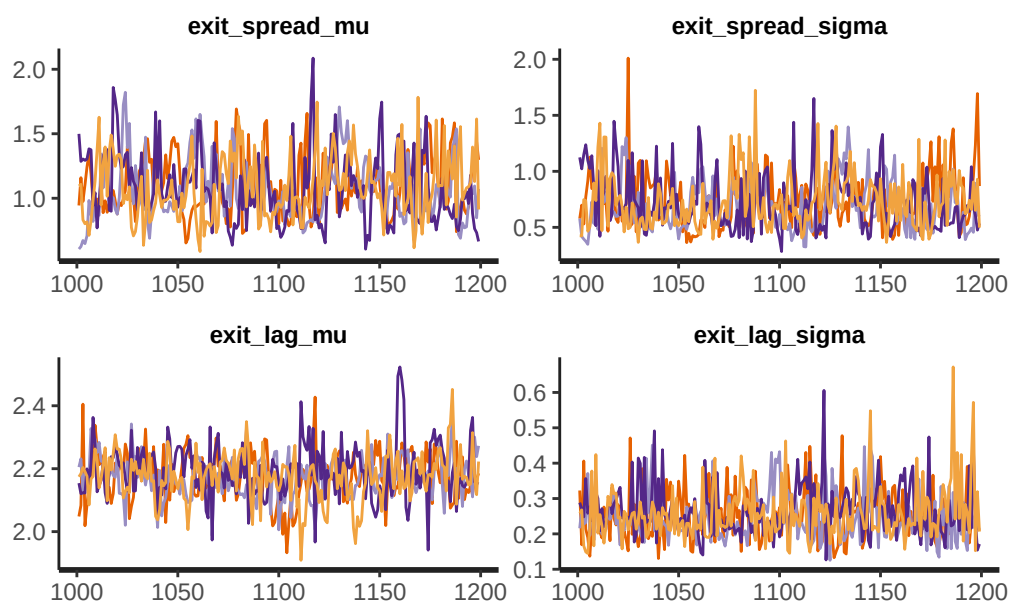


Figure 6

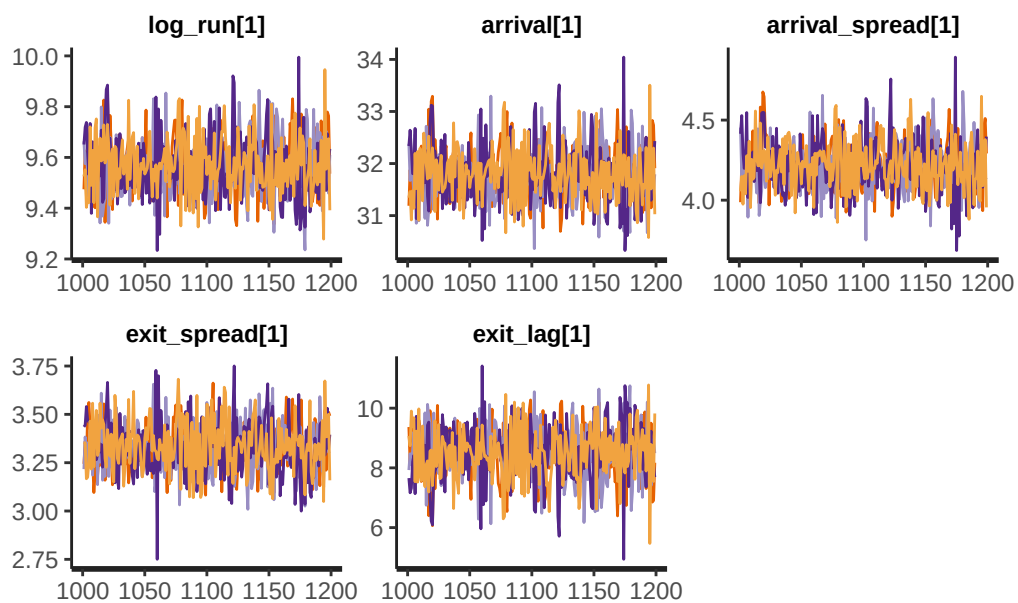
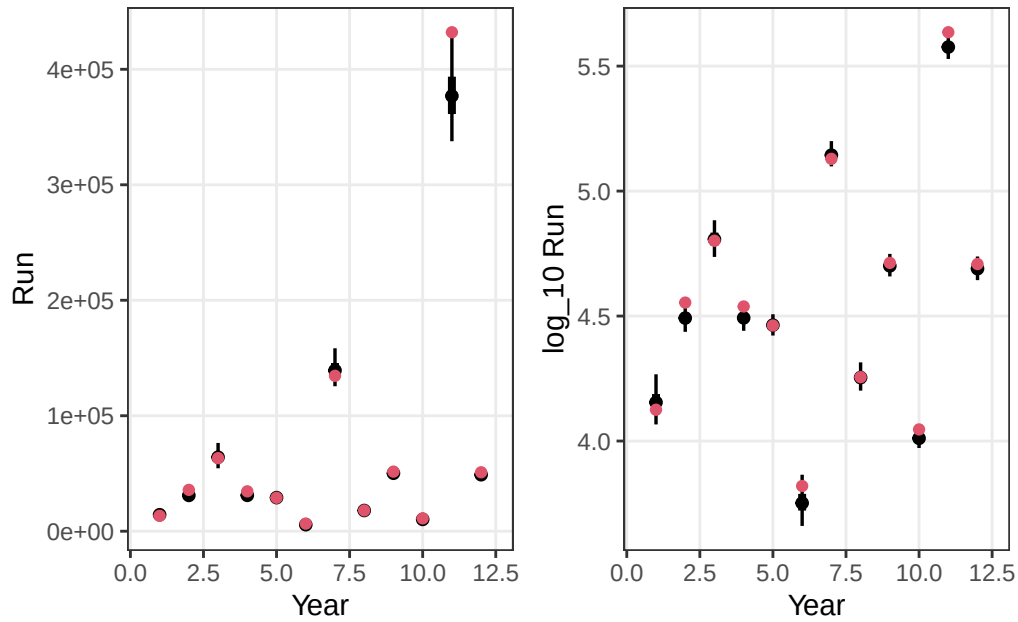
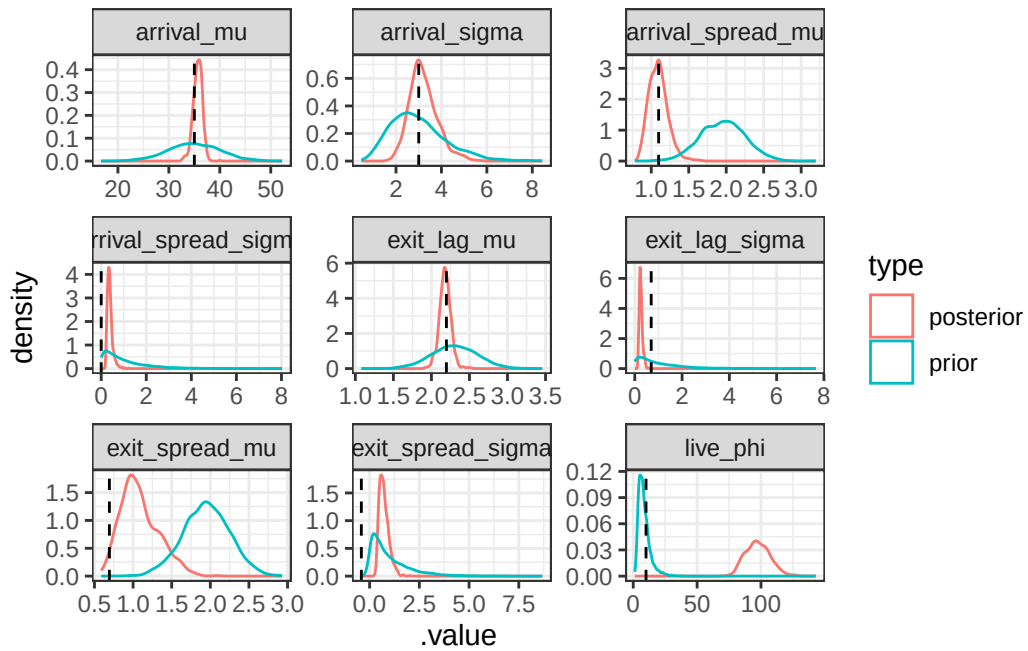


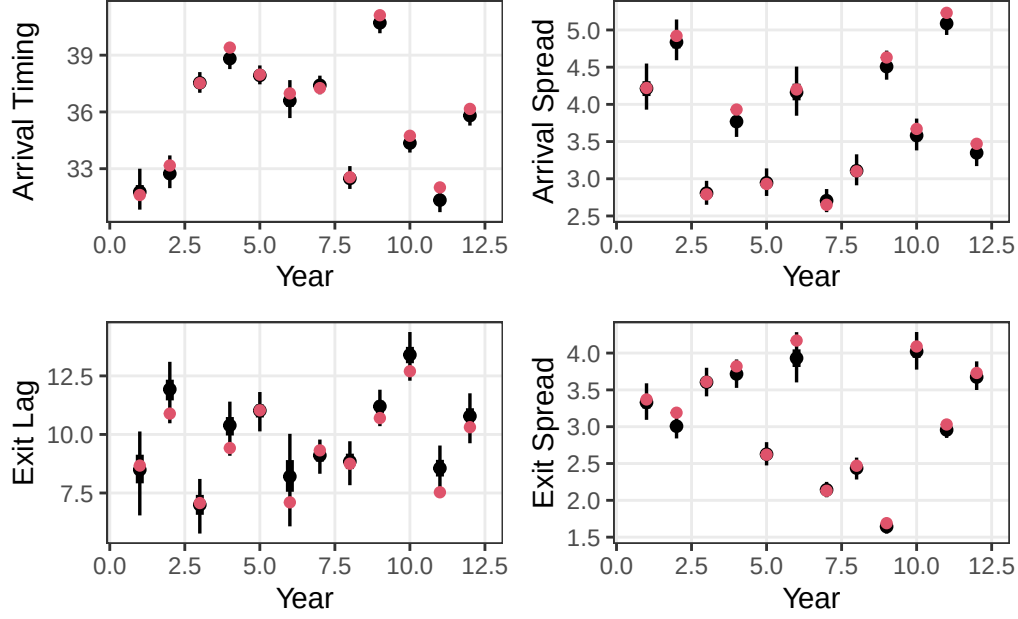
Figure 7



Fitted parameters

The annual values for spawner *arrival_timing*, *arrival_spread*, *exit_lag*, and *exit_spread* are presented as marginal posterior distributions (confidence limits) and compared to the simulated value. The goal is finding any consistent bias.





Spawner presence

The pattern of spawner distribution is recreated from the fitted parameters and presented with 95% confidence limits. This is compared to the observations with sampling error.

Stats

For comparison of scenarios of survey quality. Uses median absolute difference (*MAD*) from $\log(\text{observed}) - \log(\text{expected})$. It is faster to compute $\log(\text{observed}/\text{expected})$. The many samples representing the posterior PDD for observed are reduced to median. $\text{mad}(x)$ expands the value by 1.4826 to be the same as 1 stdev if x is a normal distribution. A mad of 0.16 is an accuracy of 16%, implying 95%CL is $\pm 2 \times \text{MAD} = 68\%$ to 132%.

Table 3: Statistics of annual parameters as ratios, each year's fitted median divided by the annual draw for that parameters before sampling error. Values for all years are summarized as median, to show bias, and mad (median absolute difference) to show precision.

Parameters	Median	MAD	CL95
Run	96%	9%	83% to 117%
Arrival Timing	99%	1%	98% to 102%
Arrival Spread	99%	2%	96% to 104%
Exit Lag	105%	8%	84% to 116%

Table 3: Statistics of annual parameters as ratios, each year’s fitted median divided by the annual draw for that parameters before sampling error. Values for all years are summarized as median, to show bias, and mad (median absolute difference) to show precision.

Parameters	Median	MAD	CL95
Exit Spread	98%	2%	97% to 103%

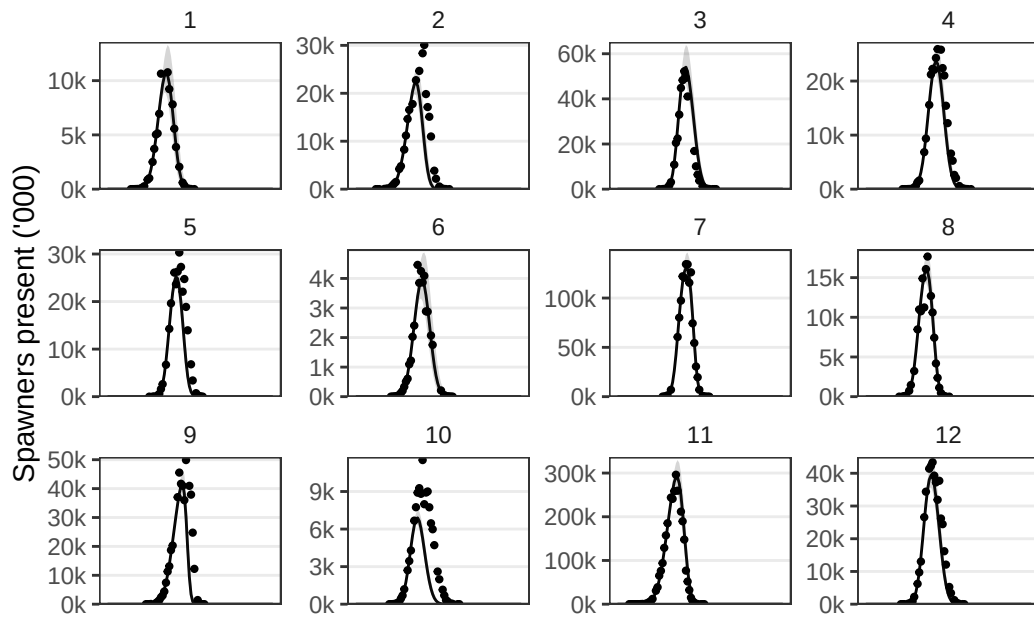


Figure 8: Estimated spawner presence, with 95% CL, from fitted parameters. Observed values are points.